FISH Glossary in WORDS Order

Organized the way the WORDS command outputs the FISH linked dictionary: from the newest definition back toward NOOP, with WORDCAT headers as section headers.
Target: stm32f407
FISH V1.8 (C)2014-2018 A-TEAM FORTH
Dictionary source: v1.8_fish-stm32f407-disco-gcc/FISH_STM32F4_FLASH.s,
v1.8_fish-stm32f407-disco-gcc/FISH_STM32F4_MEMMAP.s,
v1.8_fish-stm32f407-disco-gcc/FISH_STM32F4_SLIB.s,
v1.8_fish-stm32f407-disco-gcc/FISH_STM32F4_SOC_INIT.s,
v1.8_fish-stm32f407-disco-gcc/FISH_STM32F4_STARTUP.s,
v1.8_fish-stm32f407-disco-gcc/FISH_STM32F4_UART3_INIT.s,
v1.8_fish-stm32f407-disco-gcc/FISH_STM32F4_Forth_GNU.S,
v1.8_fish-stm32f407-disco-gcc/startup_stm32f4xx_gnu.S
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Index of Words (WORDS order)

FISH Reference Model:

FISH FISH_ONLY MYWORDS WORDS WORDCAT MS DELAY STCTR SYSCLK \ (CLS RBASE DBASE VBASE DUMP .VS .DS EHON EHOFF P PON POFF FLASH_FORGET FLASH_SAVE

INTERRUPTS:

SYSTICK_IRQ_ON SYSTICK_IRQ_OFF IRQS_SUSPEND IRQS_RESUME

FISH IO:

CR CRS ?KEY KEY EMIT TYPE ." ID. SPACE SPACES XON XOFF UART3_TX UART3_RX UART3_LSR MYBAUD UART3_INIT

NUMBER Stuff:

.D .H .B . ? U. .RU .R LSL LSR ASR REVW SXTB SXTH XOR NOT OR AND MAX MIN ABS NEGATE DECIMAL HEX BIN 0< 0= > U< < = - + */ */MOD MOD / /MOD * M/ M* 4 3 2 1 0 4- 4+ 2+ 1+ 1- 2/ 2* S->D D+ D. D.R DPL DNEGATE DABS

CREATE WORDS With:

: ; VAR VARIABLE CON CONSTANT CREATE IMMEDIATE

Inside Colon WORDS:

BEGIN WHILE REPEAT AGAIN UNTIL IF ELSE THEN ENDIF DO LOOP +LOOP <BUILDS DOES> <# # #S SIGN HOLD #> LITERAL DLITERAL [COMPILE] [ ]

Read and Write Memory With:

C! ! C@ @ +! BLANKS ERASE FILL CMOVE CLRBITS SETBITS ANDBITS

Stack Stuff:

DUP SWAP DROP OVER >R R R> LEAVE RP@ SP@ I ROT -DUP 2DUP .S .SB .SD .SH

FISH System:

COLD ABORT BYE FORGET $LEN COUNT EXPECT ' ERROR NUMBER WORD 0 ENCLOSE DIGIT ?ERROR ?ALIGNED PFA NFA CFA LFA LATEST VARALLOT C, , ALIGN ALIGNED ALLOT HERE TIB PAD IN OUT STATE CURRENT BASE DP FENCE UP BL EXECUTE NOOP

FISH Reference Model:

FISH [FISH](--)
Print Flash page status, then print the FISH sign-on message.
See FISH_ONLY, FLASH_SAVE and FLASH_FORGET

FISH_ONLY [FISH_ONLY](--)
Reset DP and UP to original power-up dictionary settings. Effectively FORGETs all ram definitions, and removes links to words saved in Flash, until COLD, BYE, FISH or a power cycle re-links them. Useful in text file download debugging.
See FISH FORGET FLASH_FORGET DP and UP

MYWORDS [MYWORDS](--)
Print only user defined words, including those saved in Flash.
See WORDS and FENCE

WORDS [WORDS](--) RENAMED --> VLIST to WORDS
List the names of all the definitions in the dictionary.
See MYWORDS

WORDCAT [WORDCAT](--)
Creates a Word Category NFA and LFA that cannot be searched for, but is displayed by WORDS and MYWORDS to label a group of Words. You can define a group of Words and then add a category name with WORDCAT. Spaces are allowed in Word Categories:
: MyAPP ;
WORDCAT MY APP WORDS:
(MY APP WORDS: will be printed at the beginning of a new line)
(MyAPP will be indented on the next line)
MYWORDS (Prints MY APP WORDS: then MyAPP)
WORDS (Prints MY APP WORDS: MyAPP and then FISH words)
WORDCAT must be on a line of it's own. The colon at the end is the FISH convention for WORDCAT's.

MS [MS](n --)
Delay n milliseconds. SYSTICK reload value is set to 1 millisecond. If you change the system clock do not use until fixed in later release. Use DELAY.
See DELAY STI_ON STI_OFF STCTR and SYSCLK

DELAY [DELAY](n value --) 0 0 DELAY \ stops timer and it's interrupts.
Delay clock cycles indicated by value, repeating n times. The ARM SYSTICK 24 bit timer is used, with an interrupt handler that increments the STCTR variable every time value countdown reaches 0. DELAY sets STCTR to negated n, and returns when STCTR reaches 0. If n=0 just initialize SYSTICK reload value.
To calculate and execute a given delay: (Do this if you change SYSCLK!)
1 SYSCLK 100d / 1 - DELAY (Value of 100 = 10 ms delay)
See MS SYSCLK STI_ON STI_OFF and STCTR

STCTR [STCTR](-- addr of STCTR)
A system variable (STCTR) containing a value incremented by the ARM SYSTICK 24bit counter interrupt handler.
See DELAY MS SYSTICK_IRQ_ON SYSTICK_IRQ_OFF and SYSCLK

SYSCLK [SYSCLK](-- value)
Return the system clock frequency in hertz.
SYSCLK .D (Prints out system clock frequency in decimal)
See DELAY MS and STCTR

**** [BACKSLASH](--)
After a trailing space treat the rest of the line as a comment.
See PAREN (

([PAREN](--) IMMEDIATE
Used in the form:
(Cccc)
Ignore comments within parenthesis. A right parenthesis, “)” ends the comments.
The comments may be multi-line but a space or tab is required for any line that is otherwise empty for formatting purposes, else a comment error will be printed. May occur during execution or in a colon-definition. A blank after the leading parenthesis is required.
(
Multiline Comments should always end the comment on it's own line.
It especially helps when commenting out code like:
1 PortC_Ack ! \ Bit Bang ACK - A closing ")" here is hard to find...
)
See \

CLS [CLS](--)
Clear the terminal display using the ANSI escape sequence ESC[2J, home the cursor, and emit CR.
See EMIT and CR

RBASE [RBASE](-- addr)
Return base addr of RAM. Do not modify values there!
See DBASE

DBASE [DBASE](-- addr)
Return base addr of user words defined in the dictionary, that are located in RAM memory until saved.
See DUMP

VBASE [VBASE](-- addr)
Return the starting address of the Ram Variable Space.
(Print the base address of the ram where variables are stored)
VBASE @ .H
See .VS and UP

DUMP [DUMP](addr n --)
Print n lines of the addr and 4 columns of 4 bytes each in hexadecimal.
addr must be a aligned by 4 or an error message will be issued. FISH is “Little Endian”, so addresses will be coherent in DUMP but character strings will be backwards in each column. Press any key to abort a DUMP.
DBASE 5 DUMP (Prints five lines of the start of the dictionary in ram.)

.VS [DOTVARSPACE](--)
Prints number of bytes available in Ram VARIABLE space in DECIMAL, not affecting BASE.
See VARALLOT, UP and .DS

.DS [DOTDICTSPACE](--)
Prints number of bytes available in dictionary in DECIMAL, not affecting BASE.
See .VS, HERE and UP

EHON [EHON](--)
Set ERROR_HALT to one. When ABORT_STOP_TILL_CO is enabled, ABORT prints the CO prompt and waits for C, O, and Enter.
See EHOFF, ERROR and ABORT

EHOFF [EHOFF](--)

Set ERROR_HALT to zero. When ABORT_STOP_TILL_CO is enabled, this skips the CO wait after an error.
See EHON, ERROR and ABORT

P [P](-- addr)
Leave the address of PROMPT on the stack. PROMPT contains zero or the address of a null-terminated prompt string.
See POFF, PON and QUIT

PON [PON](--)
Set PROMPT to the default null-terminated prompt string. QUIT resumes printing the prompt and current BASE.
See P, POFF and QUIT

POFF [POFF](--)
Set PROMPT to zero. QUIT then emits CR without the normal prompt text or BASE display.
See P, PON and QUIT

FLASH_FORGET [FLASH_FORGET](--)
Erase all Flash Pages (if any). If there are words in RAM FLASH_FORGET will not reclaim VARIABLE space or FORGET the words in Ram. Execute FISH_ONLY to reclaim VARIABLE space allocated by words in Flash if needed.
See FLASH_SAVE and FISH_ONLY

FLASH_SAVE [FLASH_SAVE](--)
Save WORDS and VARIABLE's to one of several Flash Pages if available.
A message is printed if there is nothing to save, or the last flash page has been used.
NOTE: After a power cycle the VARIABLE's saved will still be allocated, but not set to known values, so words saved in Flash must re-initialized their VARs:
VARIABLE Set? : SetVARs 4 Set? ! ; (Initialize Set? To 4)
Set? ? (Not likely to be initialized since it hasn't been initialized yet!)
(RUN will now print out message at power up)
: RUN SetVARs CR ." Set? Is now initialized to " Set? ? CR ;
FLASH_SAVE (Save Set? SetVARs and RUN to flash)
ABORT (RUN will print out message)
(Now turn power off and on and RUN will print out message)
See FLASH_FORGET and FISH_ONLY

INTERRUPTS:

SYSTICK_IRQ_ON [SYSTICK_IRQ_ON]
SYSTICK_IRQ_OFF [SYSTICK_IRQ_OFF]
IRQS_SUSPEND [CMSIS_DISABLE_IRQS]
IRQS_RESUME [CMSIS_ENABLE_IRQS]

FISH IO:

CR [CR](--)
Transmit a carriage return (ASCII 0Dh) and line feed (ASCII 0Ah) to the terminal.

CRS [CRS](n --)
Transmit n carriage returns (ASCII 0Dh) and line feeds (ASCII 0Ah) to the terminal.

?KEY [QKEY](-- f)
Return zero unless there is a char in the UART's RX FIFO – the char is not consumed.
: Wait_For_Key BEGIN ?KEY UNTIL ; (Execute until key is entered.)
Wait_For_Key (Press a key to exit this word)
KEY .H (Get the key, see and forget it.)
See KEY and UART0_LSR

KEY [KEY](-- c)
Wait for the next terminal key struck, and put the ASCII value c on the stack.

EMIT [EMIT](c --)
Transmit a ASCII character to the terminal. OUT is incremented for each character output. Character values of 0 to 0FFh are allowed versus TYPE which outputs ASCII characters from 0 to 07Fh..
FBh EMIT (FBh is a ASCII CHECKMARK symbol)
See TYPE UART0_TX UART0_LSR and OUT

TYPE [TYPE](addr count --)
EMIT count bytes in a string (only ASCII characters from 0 to 07Fh) from addr thru addr+count. OUT is incremented. EMIT allows character values from 0 to 0FFh to be output, but TYPE does not.
(Print FISH NFA, whose count has extra bits, so clear then to get count)
' FISH NFA COUNT 1Fh AND TYPE (Prints FISH NFA)
See ID. COUNT $LEN OUT and EMIT

." [DOTQ](--) IMMEDIATE
Type the quoted string, or compile an inline string to be typed at run-time. Used in the form:
." Cccc "
If executed outside a definition, ." will immediately print the text until the final " (double quote). The maximum number of characters is system dependent.

ID. [IDDOT](nfa --)
Print a dictionary name from its name field address (NFA). NFA is limited to 31 characters.
LATEST ID. (Prints the name of the last dictionary name created)
See LATEST and NFA

SPACE [SPACE](--)
EMIT a space character, ASCII 20h.

SPACES [SPACES](n --)
Transmit n ASCII blank spaces (20h) to the output device.
8 SPACES (Moves cursor 8 spaces)

XON [XON](--)
Send ASCII XON character to the terminal.
See XOFF

XOFF [XOFF](--)
Send ASCII XOFF character to the terminal.
See XON

UART3_TX [UART3_TX]

UART3_RX [UART0_RX]

UART3_LSR [UART3_LSR]

MYBAUD [MYBAUD](baudrate --)

Must be used before using UART0_INIT in the form of:
115200d MYBAUD
Because of the way FISH handles hard-fault resets, this word provides a mechanism to preserve user baud rate despite faults. UARTx_INIT, BYE or a power cycle will change the current baud rate to the baud rate specified by MYBAUD.
See UARTx_INIT

UART3_INIT [UART3_INIT]

NUMBER Stuff:

.D [DOTDEC](n --)
Print n as a number in DECIMAL, not affecting BASE.

.H [DOTHEX](n --)
Print n as a number in hexadecimal (HEX) using U. , not affecting BASE.

.B [DOTBIN](n --)
Print n as a number in binary (BIN) using U. , not affecting BASE.

. [DOT](n --) SIGNED
Print n as a number in the current BASE. A trailing blank follows. Pronounced "dot".
7 . (7)

? [QUES](addr --)
Print the value contained at the address in free format according to the current base using DOT.
Equivalent to “@ .”

U. [UDOT](value --) UNSIGNED
Print out value in current BASE as an unsigned value.
HEX -2 . (-2)
-2 U. (FFFFFFFE)
DECIMAL
See D. and .

.RU [DOTRU](n1 n2 --) UNSIGNED
Print the unsigned number n1 in the current BASE, right-aligned in a field whose width is n2. No following blank is printed.

.R [DOTR](n1 n2 --) SIGNED
Print the signed number n1 in the current BASE, right-aligned in a field whose width is n2. No following blank is printed.

LSL [LSL](n count -- n')
Logical (zero-extended) shift left by count. Valid count values are 0 to 31.
1 1 LSL . (2)
See AND OR NOT LSR and ASR

LSR [LSR](n count -- n')
Logical (zero-extended) shift right by count. Valid count values are 0 to 31.
8 1 LSR . (4)
See AND OR NOT LSL and ASR

ASR [ASR](sn count -- sn') SIGNED
Shift sn (sign-extended) right by count. Valid count values are 0 to 31.
-3 2 ASR . (-1)
See AND OR NOT LSR and LSL

REVW [REVW](n -- n)
Reverse bytes in 32bit n.

SXTB [SXB](n -- n)
Sign-extend the low byte of n to a 32-bit signed value.
See SXTH

SXTH [SXH](n -- n)

Sign-extend the low halfword of n to a 32-bit signed value.
See SXTB
SYSTICK_IRQ_OFF [SYSTICK_IRQ_OFF] (–) 0 0 DELAY is equivalent.
Turn SYSTICK Interrupt off. This will stop incrementing STCTR.
See SYSTICK_IRQ_ON DELAY MS STCTR and SYSCLK
This is not in the 812 Small Reference Model (SRM) Wordset.
SYSTICK_IRQ_ON [SYSTICK_IRQ_ON] (–) MS and DELAY turns this on.
Turn SYSTICK Interrupt on. This will resume incrementing STCTR.
See SYSTICK_IRQ_OFF DELAY MS STCTR and SYSCLK
This is not in the 812 Small Reference Model (SRM) Wordset.

XOR [XOR](xl x2 -- x)
Leave the bitwise logical exclusive-or of x1 and x2 as x.
4 6 XOR . (2)
3 0 XOR . (3)
See AND NOT ASR LSR and LSL

NOT [NOT](x1 -- x2)
Leave the bitwise logical NOT of x1 as x2.
3 NOT . (-4)
-4 NOT . (4)
See AND OR and XOR

OR [OR](x1 x2 -- x)
Leave the bitwise logical OR of x1 and x2 as x.
101b 010b OR .B (111)
1000b 101b OR .B (1101)
See AND XOR and NOT

AND [AND](x1 x2 -- x)
Leave the bit-wise logical AND of x1 and x2.
110b 10b AND .B (10)
See OR NOT ASR LSR and LSL

MAX [MAX](n1 n2 -- max) SIGNED
Leave the greater of two (signed) numbers.
-2 2 MAX . (2)

MIN [MIN](n1 n2 -- min) SIGNED
Leave the smaller of two (signed) numbers.
-2 2 MIN . (-2)

ABS [ABS](n -- u)
Leave the absolute value of n as u.
See DABS

NEGATE [NEGATE](n1 -- n2) RENAMED --> MINUS to NEGATE
Leave the Two's complement of a number.
4 NEGATE . (-4)
-4 NEGATE . (4)
See DNEGATE

DECIMAL [DECIMAL](--)
Set the numeric conversion BASE to ten. The current BASE is reflected in decimal in the system prompt.
DECIMAL (ok, go fish in BASE 10d)
See HEX and BIN

HEX [HEX](--)

Set the numeric conversion base to sixteen (hexadecimal). The current BASE is reflected in decimal in the system prompt.
HEX (ok, go fish in BASE 16d)
See DECIMAL and BIN

BIN [BIN](--)

Set the numeric conversion base to two (binary). The current BASE is reflected in decimal in the system prompt. 0 and 1 are the only valid digits in binary.
BIN (ok, go fish in BASE 2d)
See DIGIT HEX and DECIMAL.

0< [ZLESS](n -- f)

Leave a true flag if the number is less than zero (negative), otherwise leave a false flag.
1 0< . (0) -1 0< . (1)

0= [ZEQU](n -- f)

Leave a true flag is the number is equal to zero, otherwise leave a false flag.
1 0= . (0) 0 0= . (1)

> [GREATERTHAN](n1 n2 -- f) SIGNED

Leave a true flag if n1 is greater than n2 otherwise a false flag.
-1 1 > . (0) 1 -1 > . (1)

U< [ULESSTHAN](u1 u2 -- f) UNSIGNED

Leave a true flag if u1 is lower than u2, otherwise leave a false flag.
-1 -2 U< . (0)
-1 2 U< . (0)
1 2 U< . (1)
See <

< [LESSTHAN](n1 n2 -- tf/ff) SIGNED

Leave a true flag if n1 is less than n2 otherwise leave a false flag.
1 80 < . (1) 80 1 < . (0)

= [EQUAL](n1 n2 -- f)

Leave a non-zero true flag if n1=n2; otherwise leave a false flag.
2 1 = . (0) 3 3 = . (1)

- [SUB](n1 n2 -- n)

Leave the difference of n1-n2 as n.
1 2 - . (-1) 2 1 - . (1)

+ [PLUS](n1 n2 -- n) SIGNED

Add signed n1 and signed n2, leaving the signed sum.
4 3 + . (7)

***/** [SSLASH](n1 n2 n3 -- n)

Leave the ratio n = n1*n2/n3 where all are signed numbers. Retention of an intermediate 64-bit product permits greater accuracy than would be available with the sequence n1 n2 * n3 /
3 2 1 */ . (6)

***/MOD** [SSMOD](n1 n2 n3 -- rem quo)

Leave the quotient and remainder of the operation n1*n2/n3. A 64-bit intermediate product is used, as it is for */.
3 2 1 */MOD . (0 6)

MOD [MOD](n1 n2 -- modulus)

Leave the remainder of n1/n2, with the same sign as n1.
2 3 MOD . (2)
2 2 MOD . (0)
-2 3 MOD . (-2)
2 -3 MOD . (2)

/ [SLASH](n1 n2 -- n) SIGNED

Leave the signed quotient n of n1/n2.
4 2 / . (2)

/MOD [SLMOD](n1 n2 -- rem quo) SIGNED

Leave the remainder and signed quotient of n1/n2. The remainder has the sign of the dividend.
4 2 /MOD . (0 2)
0 1 2 3 4 [ZERO ONE TWO THREE FOUR] (-- n)
These small numbers are used so often that is attractive to speed up FISH by defining them by name in the dictionary as constants.

***** [STAR](n1 n2 -- n)

Leave the signed product n (32bit) of two (32bit) signed numbers n1 and n2. If the product exceeds 32 bits, it will be truncated.
2 3 * . (6)

M/ [MSLASH](d1 n1 -- srem32 squo32) SIGNED

A mixed-precision math operator which leaves the signed remainder and signed quotient from double number d1, and divisor n1.
NOTE: Result is not a Double Number but is two signed 32bit integers.
NOTE: Double numbers are created by placement of a period in the number.
3. 2 M/ . . (1 1)
-3. 2 M/ . . (-1 -1)
3. -2 M/ . . (-1 1)
See / and /MOD

M* [MSTAR](n1 n2 -- d) SIGNED

A mixed-precision math operation which leaves the double number signed product of two signed single numbers.
2 2 M* D. (4)
-2 2 M* D. (-4)
2 -2 M* D. (-4)
See *

4 [FOUR]

3 [THREE]

2 [TWO]

1 [ONE]

0 [ZERO]

4- [FOURM](n -- n-4)

8 4- . (4)

4+ [FOURP](n -- n+4)

4 4+ . (8)

2+ [TWOP](n -- n+2)

Add two to the top of the stack.

1 2+ . (3)

1+ [ONEP](n -- n+1)

Add 1 to the top of the stack.
1 1+ . (2)

1- [ONEM](n -- n-1)

Decrement n by one.
See 1+

2/ [TWOSLASH](n -- n/1)

32bit arithmetic-shift-right by one bit. Usually used as a “floored” signed divide-by-two.
4 2/ . (2)

2* [TWOSTAR](n -- n*2)

32bit left-shift by one bit position. (multiply by 2).
4 2* . (8)

S->D [STOD](n -- d) SIGNED

Sign-extend a single number, leaving a signed double number.
Could be defined as : S->D DUP 0< NEGATE ;
1 S->D D. (1)
-1 S->D D. (-1)
See D+ D.R DABS and DNEGATE

D+ [DPLUS](d1 d2 -- d) SIGNED

Leave the signed double number sum of two signed double numbers.

D. [DDOT](d --)

Print a signed double number from a 64-bit value. The high-order 32 bits are topmost on the stack.
Conversion is performed according to the current BASE. A blank space follows.
Pronounced D-dot.
See DOT

D.R [DDOTR](d n --) SIGNED

Print a signed double number d right-aligned in a field n characters wide.

DPL [DPL_SV](-- addr of NDPL)

A system variable (NDPL) containing the number of digits to the right of the decimal-point of the last double number input. It may also be used to hold the output column location of a decimal-point in user-generated formatting. The default value on single number input is -1.

DNEGATE [DNEGATE](d1 -- d2) RENAMED --> DMINUS to DNEGATE

Convert d1 to its two's-complement d2.
1. DNEGATE D. (-1)
-1. DNEGATE D. (0)
See NEGATE

DABS [DABS](d -- ud)

Leave the absolute value ud of a signed double number.
See ABS

CREATE WORDS With:

: [COLON](--) IMMEDIATE

If the dictionary is full, print an error message and abort.
Used in the form called a colon definition:
: Cccc ... ;
Creates a dictionary entry using Cccc as the word name. The definition of the word '...' is compiled into the word until ; is encountered. The compiling process is done by the text interpreter while STATE is 0xC0. IMMEDIATE words with the precedence bit P set are executed rather than being compiled.
See .DS IMMEDIATE and WORDS, which shows a category of FISH words that can be used inside colon definitions.

; [SEMI](--) IMMEDIATE

Terminate compilation of a COLON (:) definition and return STATE to zero, the interpretation STATE.
See :

VAR [VAR](--)

Create a named variable. Executing the created word leaves its storage address. VARIABLE is the high-level synonym.
See VARIABLE, VARALLOT and UP

VARIABLE [VARIABLE](--) VARIABLE and VAR defined.

A defining word used in the form:
VARIABLE Cccc
When VARIABLE is executed, it creates the definition Cccc with its cell space un-initialized. When Cccc is executed, its cell space address is put on the stack, and then used by @ or !. If the memory space for variables is full, VARIABLE will print an error message and ABORT.
NOTE: VARIABLE's cell space is allocated in RAM.
VARIABLE MyVar (Create a variable)
7 MyVar ! (Initialize MyVar to 7)
MyVar @ . (7)
See .VS UP VBASE and VARALLOT

CON [CON](n --)

Create a named constant whose execution returns n. CONSTANT is the high-level synonym.
See CONSTANT, CREATE and COMMA

CONSTANT [CONSTANT](n --) CONSTANT and CON defined.

A CONSTANT-defining word used in the form:
n CONSTANT Cccc
to create word Cccc, with its parameter field containing n. When Cccc is later executed, it will push the value of n to the stack.
8 CONSTANT EIGHT EIGHT . (8)
See VARIABLE

CREATE [CREATE](--) Maximum word length is 31 characters.

A defining word used in the form:
CREATE Cccc
Create a word in the Dictionary named Cccc. Executing the created word returns the address following itself, which is HERE. The returned address points to nothing until some companion data or code is made to be the object of the address. IF ALLOT is used after CREATE, HERE points to after the allotted memory.
If the dictionary is full, then CREATE will print an error message and ABORT.

CREATE Cccc (Cccc is now a Word, like a CONSTANT with no value.)
HERE .H (Show next available dictionary address)
Cccc .H (Will be the same – where you can put data or code)
80d ALLOT (Allocate 80 Bytes. Cccc returns the beginning of the array)
Use with DOES> inside a definition.
See <BUILDS DOES> LATEST DP CURRENT and HERE

IMMEDIATE [IMMED](--)

Occurs after a colon-definition in form:
: Cccc ; IMMEDIATE
Mark the most recently made definition (Cccc) IMMEDIATE so that when encountered at compile time, it will be executed rather than being compiled. This method allows definitions to handle unusual compiling situations, rather than build them into the fundamental compiler. The user may force compilation of an immediate definition by preceding it with [COMPILE].
(Make Cccc an immediate word)
: Cccc CR ." I'm an immediate word!" CR ; IMMEDIATE
: watch_this Cccc FISH ; (Note Cccc executes while compiling this!)
watch_this (Note that Cccc isn't in watch_this, FISH is!)
See [COMPILE] [and]

Inside Colon WORDS:

BEGIN [BEGIN](-- addr n) IMMEDIATE

Occurs in a colon-definition in form:
BEGIN ... UNTIL
BEGIN ... AGAIN
BEGIN ... WHILE ... REPEAT
At run-time, BEGIN marks the start of a sequence that may be repetitively executed. It serves as a return point from the corresponding UNTIL, AGAIN or REPEAT. When executing UNTIL, a return to BEGIN will occur if the top of the stack is false; for AGAIN and REPEAT, a return to BEGIN always occurs.
: Wait_For_Key BEGIN ?KEY UNTIL ; (Execute until key is entered.)
Wait_For_Key (Press a key to exit this word)
KEY .H (Get the key, see and forget it.)
See AGAIN UNTIL WHILE and REPEAT

WHILE [WHILE](f --) IMMEDIATE

Occurs in a colon-definition in the form:
BEGIN ... WHILE ... REPEAT
At run-time, WHILE selects conditional execution based on boolean flag f. If f is true (non-zero), WHILE continues execution of the true part thru to REPEAT, which then branches back to BEGIN. If f is false (zero), execution skips to just after REPEAT, exiting the structure.
: BeginTest (0 1 --) BEGIN ." Begin " WHILE ." While " REPEAT CR ;
(The argument are successively used by WHILE)
0 1 BeginTest (Begin While Begin)
1 0 BeginTest (Begin)
See BEGIN and REPEAT :

REPEAT [REPEAT](addr1 n1 addr2 n2 --) IMMEDIATE

Used within a colon-definition in the form:
BEGIN ... WHILE ... REPEAT
At run-time, REPEAT forces an unconditional branch back to just after the corresponding BEGIN. Compiles a branch back to BEGIN, and computes a branch offset for WHILE.
: BeginTest (0 1 --) BEGIN ." Begin " WHILE ." While " REPEAT CR ;
(The argument are successively used by WHILE)
0 1 BeginTest (Begin While Begin)
1 0 BeginTest (Begin)
See BEGIN and WHILE

AGAIN [AGAIN](--) IMMEDIATE

Used in a colon-definition in the form:
BEGIN ... AGAIN
At run-time, AGAIN forces execution to return to corresponding BEGIN. There is no effect on the stack. Execution cannot leave this loop (unless R> DROP is executed). At compile time, AGAIN compiles the :NONAME [BRAN] instruction with an offset back to BEGIN (from HERE at the time to addr).
See BEGIN for example.

UNTIL [UNTIL](f --) IMMEDIATE

Occurs within a colon-definition in the form:
BEGIN ... f UNTIL
At run-time, UNTIL controls the conditional branch back to the corresponding BEGIN. If f is false, execution returns to just after BEGIN; if true, execution continues ahead to the next word after UNTIL.

(?KEY returns zero (false flag) until a key is pressed, then a true flag)
: TestUntil BEGIN ?KEY UNTIL ; (will execute until a key is pressed }
KEY DROP (Remove the key pressed)
See BEGIN AND AGAIN

IF [IF](f --) IMMEDIATE

Occurs is a colon-definition in form:
IF ... ENDIF
IF ... ELSE ... ENDIF
At run-time, IF selects execution based on a Boolean flag. If the flag is true (non-zero), execution continues ahead thru the IF part. If flag is false (zero), execution skips till just after ELSE to execute the false part. After either part, execution resumes after ENDIF. ELSE and its false part are optional; if ELSE is missing, false execution skips to just after ENDIF.
: True? IF ." YES" ELSE ." NO" ENDIF ;
1 True? (Prints YES)
0 True? (Prints NO)
See also ELSE ENDIF and THEN

ELSE [ELSE](--) IMMEDIATE

Occurs within a colon-definition in the form:
IF ... ELSE ... ENDIF
At run-time, the ELSE section of words until ENDIF are executed if the argument to IF is false.
: True? IF ." YES" ELSE ." NO" ENDIF ;
1 True? (Prints YES)
0 True? (Prints NO)
See IF ENDIF and THEN

THEN [THEN](--) IMMEDIATE

An alias for ENDIF.
See IF ELSE and ENDIF

ENDIF [ENDIF](addr 2 --) IMMEDIATE

Occurs within a colon-definition in the form:
IF ... ENDIF
IF ... ELSE ... ENDIF
At run-time, ENDIF serves s the destination of the branch from IF or ELSE., depending on the argument passed to the IF.
: True? IF ." YES" ELSE ." NO" ENDIF ;
1 True? (Prints YES)
0 True? (Prints NO)
See also IF ELSE and THEN

DO [DO](n1 n2 --) IMMEDIATE

Occurs in a colon-definition in the form:
n1 n2 DO ... LOOP
n1 n2 DO ... +LOOP
At run-time, DO begins a sequence with repetitive execution controlled by a loop limit n1 and an index with initial value n2. Upon reaching LOOP the index is incremented by one. +LOOP is incremented by another inline index argument, explained in the example below. Until the updated index equals or exceeds the limit, execution loops back to just after DO. When the index is equal or exceeded execution continues after the LOOP or +LOOP. Within a loop, I will copy the current value of the loop-index to the stack for use as variable in the construct.
: print10 10 0 DO I . LOOP ;
print10 (0 1 2 3 4 5 6 7 8 9)
: print10+loop 10 0 DO I . 2 +LOOP ;

print10+loop (0 2 4 6 8)
See I LOOP +LOOP and LEAVE

LOOP [LOOP](addr n --) IMMEDIATE

Occurs in a colon-definition in the form:
n1 n2 DO ... LOOP
At run-time, LOOP selectively controls branching back to the corresponding DO based on the loop index and limit. The loop index is incremented by one and compared to the limit. The branch back to DO occurs until the index equals or exceeds the limit; at that time, the parameters are discarded and execution continues ahead.
: print10 10 0 DO I . LOOP ;
print10 (0 1 2 3 4 5 6 7 8 9)
See DO I +LOOP and LEAVE

+LOOP [PLOOP](n --) IMMEDIATE

Occurs in a colon-definition in the form:
n1 n2 DO ... +LOOP
At run-time, +LOOP selectively controls branching back to the corresponding DO based on n, the loop index, and the loop limit. The signed increment n is added to the index and the total compared to the limit.
The branch back to DO occurs until the new index is equal to or greater than the limit (n1>0), or until the new index is equal to or less than the limit (n1<0). Upon exiting the loop, the parameters are discarded and execution continues ahead.
: print10+loop 10 0 DO I . 2 +LOOP ;
print10+loop (0 2 4 6 8)
See DO I LOOP and LEAVE

<BUILDS [BUILDS](--) CREATE can be used instead.

Used within a colon definition:
: Cccc <BUILDS ... DOES> ... ;
Each time Cccc is executed, <BUILDS defines a new word with a high-level execution procedure.
Executing Cccc in the form:
Cccc nnnn
uses CREATE to create a dictionary entry for nnnn with a call to the DOES> part for nnnn. When nnnn is later executed, it has the address of its parameter area on the stack and executes the words after DOES> in Cccc. <BUILDS and DOES> allow run-time procedures to written in high-level rather than in assembler code.
: DCONST <BUILDS , , DOES> DUP @ . 4 + @ . ;
-1 -1 DCONST DNEGATIVEONE
DNEGATIVEONE D. (-1)
See DOES> and CREATE

DOES> [DOES](pfa --)

Occurs in a colon-definition in the form:
CREATE ... DOES> … \ ANS
<BUILDS ... DOES> … \ F83
This construct is used to make other words that execute the words you place after the DOES>. Used in combination with CREATE or <BUILDS. When the DOES> part executes, it begins with the address of the first parameter of the new word on the stack. This allows interpretation using this area, or its contents. Typical uses include assemblers, multidimensional arrays, and compiler generation.
(Create a word that creates constants that print their value.)
: .CONST CREATE , DOES> @ . ;
(7 is compiled in the new word by the comma above)
7 .CONST SEVEN
(The words after DOES> execute with the PFA of seven on the stack)

SEVEN (7)
See <BUILDS and CREATE

<# [BDIGS](--)
Setup for pictured numeric output formatting using the words:
<# # #S SIGN #>
The conversion is done on a double number producing text at PAD.
(In this example 24h is ASCII $ and 2Eh is . <Period>)
: .DOLLARS <# # # 2Eh HOLD (no prompt until this I finished!)
#S 24h HOLD #> TYPE ; (now you will see a prompt!)
56668. .DOLLARS ($566.68)
See # #S #> HOLD and SIGN

[DIG](d1 -- d2)
Occurs in a colon-definition in the form:
<# # n HOLD #S SIGN #>
Generate from a double number d1, the next ASCII character which is placed in an output string, usually in PAD. Result d2 is the quotient after division by BASE, and is maintained for further processing. Used between <# and #>. Each number = BASE_SV*number+(character-'0')
See #S HOLD SIGN and <# or #> for example code.

#S [DIGS](d1 -- d2)
Occurs in a colon-definition in the form:
<# # n HOLD #S SIGN #>
Generates ASCII text in the text output buffer, usually PAD, by the use of #, until a zero double number d2 results. Used with #, HOLD, and SIGN between <# and #>.
See # HOLD SIGN <# and #> for example code.

SIGN [SIGN](n d -- d)
Occurs in a colon-definition in the form:
<# # n HOLD #S SIGN #>
Place an ASCII "-" sign just before a converted numeric output string in the text output buffer when n is negative. n is discarded but double number d is maintained. Must be used between <# and #>.
(In this example 24h is ASCII $ and 2Eh is . <Period>)
: .SIGNEDDOLLARS (n d --)
<# # # 2Eh HOLD #S 24h HOLD SIGN #> TYPE ; (PRESS ENTER)
-1 56668. .SIGNEDDOLLARS (-$566.68)
See <# # #S #> and HOLD

HOLD [HOLD](c --)
Occurs in a colon-definition in the form:
<# # n HOLD #S SIGN #>
Used in colon definition between <# and #> to insert an ASCII character into a pictured numeric output string.
2Eh HOLD (will create a decimal point in the output number.)
See SIGN # #S #> HOLD and <# for example code

#> [EDIGS](d -- addr count)
Occurs in a colon-definition in the form:
<# # n HOLD #S SIGN #>
Terminates numeric output conversion, leaving the text address and character count suitable for use with TYPE.
(In this example 24h is ASCII $ and 2Eh is . <Period>)
: .DOLLARS <# # # 2Eh HOLD (CR - no prompt until this is finished!)
#S 24h HOLD #> TYPE ; (now you will see a prompt!)
56668. .DOLLARS ($566.68)

See <# # #S HOLD and SIGN

LITERAL [LITERAL](--) IMMEDIATE

If compiling, then compile a 32bit literal created within a definition, in the form:
: Cccc [(create 32bit value] LITERAL ;
Does nothing if interpreted. This definition is IMMEDIATE so that it will execute during a colon definition. This means the compilation must be suspended by [, allowing for compile-time calculation of the value, then resumed compiling by] , and then LITERAL will compiles this value into the definition.
: Cccc [3 2 *] LITERAL ; (Cccc computes 6 while compiling!)
Cccc . (6)
See DLITERAL IMMEDIATE [COMPILE] [and]

DLITERAL [DLITERAL](-- d) IMMEDIATE

If compiling, then compile a 64-bit literal created within a definition, in the form:
: Cccc [(create 64bit value] DLITERAL ;
Does nothing if interpreted. This definition is IMMEDIATE so that it will execute during a colon definition. This means the compilation must be suspended by [, allowing for compile-time calculation of the value, the resume compiling by] , and LITERAL compiles this value in the definition.
: Cccc [3. 2. D+] DLITERAL ; (Cccc computes 5 while compiling!)
Cccc D. (5)
See LITERAL IMMEDIATE [COMPILE] [and]

[COMPILE] [BCOMP](--) IMMEDIATE

Compile next word in definition even if it is an IMMEDIATE word that would otherwise execute during compilation. Used in a colon-definition in the form:
: Cccc [COMPILE] FISH ;
Cccc (Acts like FISH i.e. Prints what FISH prints)
See [and]

[[LBRAC](--)

Used in a colon-definition in form:
: xxx [words] more ;
Suspend compilation by setting STATE to zero (0C0h is compiling). The words after [are executed, not compiled. This allows calculation or compilation exceptions before resuming compilation with] .
: Cccc [3 2 *] LITERAL ; (Cccc computes 6 while compiling!)
Cccc . (6)
See STATE LITERAL and]

] [RBRAC](--) IMMEDIATE

Resume compilation, by setting STATE back to 0C0h (zero is interpreting). Setting STATE to 0C0h is used in INTER to execute IMMEDIATE words while compiling.
: Cccc [3 2 *] LITERAL ; (Cccc computes 6 while compiling!)
Cccc . (6)
See [and STATE

Read and Write Memory With:

C! [CSTORE](n addr --)
Store the 8-bit byte in the 8 least significant-bits of n at addr.
See C@

! [STORE](n addr --)
Write 32bit n to addr. Pronounced "store".
See C!

C@ [CAT](addr -- b)
Leave the 8-bit contents of addr on the stack as a 32 but number.
Pronounced “C-Fetch” or “Char Fetch”.
See C!

@ [AT](addr -- n)
Get the 32bit contents of addr and place the contents on TOS.
Pronounced “fetch”.

+! [PSTORE](u addr --)
Add u to the value at the address. u can be positive or negative.
Pronounced "plus-store".

BLANKS [BLANKS](addr count --)
Fill an area of memory beginning at addr with blanks (20h).
See ERASE FILL PAD and DUMP

ERASE [ERASE](addr n --)
Clear a region of memory to zero from addr to addr+n.
PAD 8 ERASE (Fill 8 bytes of PAD with zero)
PAD 2 DUMP (Display 2 lines of PAD with 8 zeros)
See FILL PAD and DUMP

FILL [FILL](addr quan b --)
Fill memory at the address with the specified quantity of bytes b.
PAD 8 33h FILL (Fill 8 bytes of PAD with 33h)
PAD 2 DUMP (Display 2 lines of PAD with 8 33h's)
See ERASE PAD and DUMP

CMOVE [CMOVE](addr1 addr2 count --)
Move the specified count of bytes beginning at addr1 to addr2. The contents of from_addr is moved first, proceeding toward high memory.
See FILL

CLRBITS [CLRBITS]

SETBITS [SETBITS](addr val --)
Logical OR the value-bits with the contents of addr using read, modify, write method.
(Create and initialize variable TEST_SETBITS)
VARIABLE TEST_SETBITS 110b TEST_SETBITS !
(Use SETBITS to modify TEST_SETBITS)
TEST_SETBITS 10001b SETBITS
(View results)
TEST_SETBITS @ .B (10111)
See CLRBITS and ANDBITS

ANDBITS [ANDBITS](addr value --)

Logical AND the value-bits with the contents of addr using read, modify, write method.

```
( Create and initialize variable TEST_ANDBITS )
VARIABLE TEST_ANDBITS 110b TEST_ANDBITS !
( Use ANDBITS to modify TEST_ANDBITS )
TEST_ANDBITS 10b ANDBITS
( View results )
TEST_ANDBITS @ .B ( 10 )
```

See CLRBITS and SETBITS

Stack Stuff:

DUP [DUP](x -- x x)

Duplicate the item on the top of the stack.
See DROP SWAP ROT and OVER

SWAP [SWAP](xl x2 -- x2 x1)

Exchange the top two items on the stack.
1 2 .S (TOS> 2 1)
SWAP .S (TOS> 1 2)
See DUP DROP ROT and OVER

DROP [DROP](n --)

Discard n from the top of the stack.
See DUP SWAP ROT and OVER

OVER [OVER](nl n2 -- nl n2 n1)

Copy the second stack value, placing it as the new TOS.
1 2 .S (TOS> 2 1)
OVER .S (TOS> 1 2 1) DROP DROP DROP (Remove example items)
See DUP SWAP ROT and OVER

>R [TOR](x --) R:(–- x)

Remove an item from the computation stack and place it on top of the return stack. Use must be balanced with R> in the same definition. Pronounced to-r.
(In this example 3 would print 3 times if >R and R> weren't used)
: ShowRstackUsage 3 DUP . >R .S R> . ;
ShowRstackUsage (3 3)
See R> and R

R [R](-- x)

Copy the top of the return stack to the parameter stack. Can only be used inside a colon definition. Use must be balanced with >R in the same definition.
(In this example 3 is printed 3 times.)
: ShowRstackUsage 3 DUP . >R R . R> . ;
ShowRstackUsage (3 3 3)
See >R and R>

R> [RFROM](-- x)

Remove the top item from the return stack and leave it on the parameter stack. Can only be used inside a colon definition. Pronounced r-from.
See >R and R

LEAVE [LEAVE](--)

Force termination of a DO loop the next time LOOP or +LOOP is executed.
(This example will leave the loop after the loop index = 4)
: NotAllLoop 10 0 DO I . I 4 = IF LEAVE THEN LOOP ;
NotAllLoop (0 1 2 3 4)

RP@ [RPAT](-- addr)

Place the address of the top of the return stack on the parameter stack, as it was before RP@ was executed.

SP@ [SPAT](-- addr)

Return the address of the current stack position to the top of the stack, as it was before SP@ was executed.

I [I](-- n)

Copy the value on top of the return stack to the parameter stack. Used within a DO LOOP and DO +LOOP to copy the current loop index to the stack.
See R DO LOOP +LOOP

ROT [ROT](nl n2 n3 -- n2 n3 nl)

Rotate the top three values on the stack, bringing the third to the top.
1 2 3 .S (TOS> 3 2 1)
ROT .S (TOS> 1 3 2) DROP DROP DROP (Remove example items)
See .S SWAP OVER DUP and DROP

-DUP [ZNDUP](n -- n) (if zero)

(n -- n n) (non-zero)
Reproduce n only if it is non-zero. This is usually used to copy a value just before IF, to eliminate the need for an ELSE part to drop it.

2DUP [TDUP](n2 n1 -- n2 n1 n2 n1)

Duplicate top two stack items. The prefix of 2 convention means an operation on the top two stack items.
See -DUP

.S [DOTS](--)

Non-destructively print items on stack in current BASE. TOS> is printed first.

.SB [DOTSBIN](--)

Non-destructively print items on stack in binary (BIN), not affecting BASE. TOS> is printed first.

.SD [DOTSDEC](--)

Non-destructively print items on stack in DECIMAL, not affecting BASE. TOS> is printed first.

.SH [DOTSHEX](--)

Non-destructively print items on stack in hexadecimal (HEX), preserving BASE. TOS> is printed first.

FISH System:

COLD [COLD](--)

Reset FISH system and execute ABORT, which execute RUN if saved in Flash, else restarts the outer interpreter, QUIT.
>>> CAUTION: all user words in RAM are forgotten.
See ABORT BYE and FISH_ONLY

ABORT [ABORT](--)

Clear the stacks and enter the execution state. Does not reset the dictionary. If RUN is defined in Ram or Flash execute it first.
See COLD and BYE

BYE [BYE](--)

Cold-restart with serial initialization in last baud rate set by MYBAUD. If MYBAUD not used the deafult baud rate will be used. Executes ABORT
See COLD and ABORT.

FORGET [FORGET](--)

Executed in the form:
FORGET Cccc
Deletes the definition named Cccc from the dictionary in RAM, along with all entries that physically follow it. VARIABLE space is reclaimed. Prints .VS and .DS.. NOTE: Cannot be used to forget words in Flash.
See FENCE .VS .DS and FLASH_FORGET

$LEN [NULLSTRLEN](addr -- addr len)

Count length of null terminated string, leaving addr and len on stack.
Can be used with TYPE.
PAD 8 EXPECT (waits for user to type 8 characters)
PAD $LEN TYPE
See TYPE COUNT and PAD

COUNT [COUNT](addr1 -- addr2 n)

The byte at addr1 contains the count of the text starting at addr+1. COUNT leaves the byte-count n and the text address addr2 (addr1+1). COUNT is usually followed by TYPE. User Strings may be up to 255 characters.
See $LEN and TYPE

EXPECT [EXPECT](addr count --)

Wait for user input, characters from the terminal to the buffer starting at addr+1, until a "return" or the count of characters have been received. One or more nulls are added at the end of the text creating a null-terminated string. A count byte is placed at addr, the count being derived by $LEN. The count byte limits strings to 255 characters long. The backspace and Delete key erases previous character making EXPECT a useful TextBox primitive. Use COUNT with TYPE for strings saved this way. $LEN can be used with addr+1.
PAD 8 EXPECT (waits for user to type 8 characters or CR is entered)
PAD COUNT TYPE (Print what you typed that was saved in PAD)
See PAD $LEN COUNT NUMBER and TYPE

' [TICK](-- pfa) IMMEDIATE

Used in the form:
' Cccc
When interpreting leaves the parameter field address of the dictionary word Cccc. If the word is not found after a search of CURRENT, an appropriate error message is given. When compiling it is

executed inside a colon-definition because it is an IMMEDIATE word. Used to compile the address as a literal.
' FISH CFA EXECUTE (Execute FISH, same as typing FISH)
' WORDS NFA ID. (Print out the name field of WORDS)
: Cccc ' FISH CFA EXECUTE ; (Cccc executes FISH)
Pronounced "tick".
See CFA NFA LFA ID. and IMMEDIATE

ERROR [ERROR](cntstr --)

Notify the user of a fatal error by printing the counted string, then execute ABORT to restart the system. With ABORT_STOP_TILL_CO enabled and ERROR_HALT nonzero, ABORT displays the CO prompt and waits for C, O, and Enter before returning to QUIT.
See ABORT and ?COMP

NUMBER [NUMBER](addr -- d)

Convert a counted character string at addr (with a null following) to a signed double number, using the current numeric base. If a decimal point is encountered, its position will be given in DPL, but no other effect occurs. If numeric conversion is not possible, an error message will be given.
PAD 3 EXPECT (<cr> enter 123)
PAD NUMBER D. (123)
See WORD

WORD [WORD](c --)

Parse the text in TIB , until a delimiter c is found. Move the character string to HERE, with a count byte and 2 nulls at the end. Leading occurrences of c are ignored. IN is incremented (IN is initialized to 1 to skip count byte). Usually used inside a definition.
: UseWORD 0Dh WORD CR .” Your word is “ HERE COUNT TYPE CR ;
UseWORD MyWord! (Prints Your word is MyWord!)
See IN

0 [NULL](--) IMMEDIATE

Executed by Enter or when input line length maximum is encountered in the INTER -pret loop. Returns to QUIT loop.
See TIB and EXPECT

ENCLOSE [ENCL](addr1 c -- addr1 n1 n2 n3)

The text scanning primitive used by WORD. From the text address addr1 and an ASCII delimiting character c, is determined the byte offset to the first non-delimiter character n1, the offset to the first delimiter after the text n2, and the offset to the first character past the enclosed text. This procedure will not process past an ASCII 'NULL', treating it as an unconditional delimiter. NULL terminated strings are provided by EXPECT.
See EXPECT

DIGIT [DIGIT](c base -- n 1) (ok)

(c base -- 0 (bad)
Converts the ASCII character c (using BASE, c must be valid in BASE) to its binary equivalent n, accompanied by a true flag. If the conversion is invalid, DIGIT leaves only a false flag.

?ERROR [QERROR](f cntstr --)

If f is true, then TYPE error message (cntstr) and ABORT.

?ALIGNED [QALIGNED](value --)

Abort with a diagnostic if value is not aligned to a 4-byte boundary. The value is consumed.
See ALIGNED, ABORT and ERROR

PFA [PFA](nfa -- pfa)

Convert the Name Field Address (NFA) of a compiled definition to its Parameter Field Address (

PFA)
. (' FISH leaves the PFA of FISH on the stack)
' FISH CFA EXECUTE (types out the sign-on message)
See CFA and LFA

NFA [NFA](pfa -- nfa)

Convert the parameter field address (PFA) of a definition to its name field address (NFA).
(' FISH leaves the PFA of FISH on the stack)
' FISH NFA ID. (types the NFA of FISH)
See PFA CFA and LFA

CFA [CFA](pfa -- cfa')

Convert the Parameter Field Address (PFA) of a definition to its Code Field Address (CFA). CFA's are the execution entry point for words.
(' FISH leaves the PFA of FISH on the stack)
' FISH CFA EXECUTE (types out the sign-on message)
See PFA and LFA
CLRBITS [CLRBITS] (addr value)
Logical AND then NOT the value-bits with the contents of addr using read, modify, write method.
(Create and initialize variable TEST_CLRBITS)
VARIABLE TEST_CLRBITS 110b TEST_CLRBITS !
(Use ANDBITS to modify TEST_CLRBITS)
TEST_CLRBITS 10b CLRBITS
(View results)
TEST_CLRBITS @ .B (100)
See ANDBITS and SETBITS

LFA [LFA](pfa -- lfa)

Convert the parameter field address (PFA) of a word in the dictionary to its link field address (LFA).
' FISH (leaves FISH's PFA)
LFA (LFA contains the NFA of the next word in the dictionary!)
@ ID. (Get that NFA and print it!)
See PFA LFA and CFA

LATEST [LATEST](-- addr)

LATEST leaves the name field address (NFA) of the last word defined, pointed to by CURRENT.
(Print the name of the last definition created in the dictionary)
LATEST ID.
See CURRENT HERE and DP

VARALLOT [VARALLOT](n -- addr) ALIGNED

Add n * 32 bits to the systems ram variable space pointer UP. Used to allocate space in RAM for VARIABLE's and other data structures. If the memory space for variables is full, VARALLOT will print an error message and ABORT.
See .VS and VBASE

C, [CCOMMA](n --)

Store the least-significant 8-bits of n into the next available dictionary byte, advancing the dictionary pointer.
>>> CAUTION: ALLOT and C, are the only words that can allot uneven amounts.
NOTE: Use ALIGN after ALLOT or C, when alignment is needed.
See , ALIGNED and ALIGN

, [COMMA](n --) ALIGNED

Write n into next ALIGNED dictionary memory cell; DP is incremented.

See HERE and LATEST

ALIGN [ALIGN](--) DPANS94:

Pads the current DP with 0FFh until aligned. Used in , reflected by HERE.
Use when DP may be unaligned and the next value to be compiled needs to be aligned, for example after an odd amount ALLOT'ed or odd number of C, usage.
See ALIGNED C, and ALLOT

ALIGNED [ALIGNED](addr -- aligned-addr) DPANS94

Modify addr if necessary to ensure it is aligned to a 32bit boundary.
See ALIGN

ALLOT [ALLOT](n --) SIGNED

Add the signed number to the dictionary pointer DP. May be used to reserve or de-allocate dictionary space.
CAUTION! ALLOT and C, are the only words that can allot uneven amounts.
NOTE: you can use ALIGN after ALLOT or C, to insure DP is aligned.
See ALIGN ALIGNED VARALLOT DP HERE and LATEST

HERE [HERE](-- addr)

Leave the address of the next available dictionary location, stored in the system variable DP.
See DP and LATEST

TIB [TIB_SV](-- addr of TIB)

Leave the address of the system text input buffer TIB on the stack. Typically TIB is at least 82 bytes long.
See IN EXPECT and WORD

PAD [PAD_SV](-- addr of PAD)

Leave the address of PAD on the stack. PAD is typically at least 82 bytes long. PAD may be used in programs as a temporary buffer.
PAD 8 EXPECT (waits for user to type 8 characters or CR is entered)
PAD COUNT TYPE (Print what you typed that was saved in PAD)
PAD 8 33h FILL (Fill 8 bytes of PAD with 33h)
PAD 2 DUMP (Display 2 lines of PAD with 8 33h's)
See ERASE FILL EXPECT TYPE and DUMP

IN [IN_SV](-- addr of IN)

A system variable (IN) containing the byte offset within the current input text buffer (TIB) from which the next text will be accepted. WORD uses and adjusts the value of IN. IN is initialized to 1 to skip count byte.
See EXPECT TIB and OUT
IRQS_RESUME [CMSIS_ENABLE_IRQS] (–) CMSIS __enable_irq();
IRQS_SUSPEND [CMSIS_DISABLE_IRQS] (–) CMSIS __disable_irq();
These are not in the 812 Small Reference Model (SRM) Wordset.

OUT [OUT_SV](-- addr of OUT)

A system variable (OUT) that contains a value incremented by EMIT. The user may examine OUT for output display formatting, typically by EMIT-ing of further spaces to reach a certain line position to format output.

STATE [STATE_SV](-- addr of CSTATE)

A system variable (CSTATE) containing the compilation state. A non-zero value indicates compilation. The value itself will be 0C0h while compiling and zero when interpreting. 0C0h must be used to execute IMMEDIATE words while compiling colon definitions.

CURRENT [CURRENT_SV](-- addr of CURRENT)

Leave the address of the system variable (CURRENT), which contains the address of the NFA of the last word in the dictionary. CURRENT is searched every time the interpreter looks for a word.
CURRENT ? LATEST . (2 numbers printed that are the same)
See LATEST

BASE [BASE_SV](-- addr)

A system variable containing the current number base used for input and output conversion. The current BASE is reflected in decimal in the system prompt.
See BIN DECIMAL HEX and DIGIT

DP [DP_SV](-- addr of DP)

A system variable, the dictionary pointer, which contains the address of the next free memory in the dictionary. The value may be read by HERE.
HERE .H (Prints the address in DP in hexadecimal)
See : , ALLOT and CALLOT

FENCE [FENCE_SV](-- addr of FENCE)

A system variable (FENCE) containing an address below which FORGETting is trapped. To forget below this point the user must alter the contents of FENCE.
See FORGET MYWORDS and FLASH_FORGET

UP [UP_SV](-- addr of UP) ALIGNED

A system variable (UP), the VARIABLE and VARALLOT pointer, which contains the address of the next free memory available for VARIABLE and VARALLOT. The value is read by .VS and altered by VARIABLE and VARALLOT.
See .VS

BL [BLANK](-- n)

Push ASCII space character (20h) to TOS.

EXECUTE [EXECUTE](cfa --) RENAMED--> EXECUTE to EXECUTE

Execute the definition whose Code Field Address (CFA) is on the stack. The
code field address is also called the compilation address.
' WORDS CFA EXECUTE (execute WORDS, same as typing WORDS)
See the Symbol file for all NFAs and CFAs in FISH, and the :NONAME section of this document whose names are CFA labels in the symbol file.
See PFA and CFA

NOOP [NOOP](--)

A do nothing word that performs no operation.

Appendix: Glossary Entries Not in This Dictionary

?ADDR [QADDR](addr --)

If addr not aligned by 4 bytes, then issue message and ABORT.

UARTx_INIT [UARTx_INIT](--) x is board dependent.

Change Baud Rate of the terminal according to value set with MYBAUD, using 8 bit characters, no parity and 1 Stop Bit, with the software flow control mechanism named XON XOFF. To change the baud rate, MYBAUD must be used first. This is to preserve your baud during any reset. UARTx_INIT will set the baud rate to the system DEFAULT_BAUD unless MYBAUD is used. On STM32F407, the concrete dictionary name is UART3_INIT.
See MYBAUD

UARTx_RX [UARTx_RX](-- addr) x is board dependent.

UART RX register. This is the serial terminal character input register. On STM32F407 this is UART3_RX, the USART3 receive-data register address.
See UARTx_RX and UARTx_LSR

UARTx_TX [UARTx_TX](-- addr) x is board dependent.

UART TX register. This is the serial terminal character output register. On STM32F407 this is UART3_TX, the USART3 transmit-data register address.
See UARTx_RX and UARTx_LSR

UARTx_LSR [UARTx_LSR](-- value) x is board dependent.

UART Line Status Register. This is the serial terminal status register. Returns bits indicating character available, ready to transmits and error conditions. On STM32F407 this is UART3_LSR and reads the USART3 status byte.
See UARTx_TX and UARTx_RX

Appendix: Undocumented Words

These dictionary words have no RM glossary entry. They are internal words documented only by the symbol file.
To use one, find its CFA in the symbol file and EXECUTE it, e.g. ' SYSTICK_IRQ_ON CFA EXECUTE .

```
SYSTICK_IRQ_ON   SYSTICK_IRQ_ON      0x08003130
SYSTICK_IRQ_OFF  SYSTICK_IRQ_OFF     0x08003104
IRQS_SUSPEND     CMSIS_DISABLE_IRQS  0x080030d8
IRQS_RESUME      CMSIS_ENABLE_IRQS   0x080030b0
UART3_TX         UART3_TX            0x08002e08
UART3_RX         UART0_RX            0x08002df0
UART3_LSR        UART3_LSR           0x08002dc0
UART3_INIT       UART3_INIT          0x08002d24
4                FOUR                0x080026e8
3                THREE               0x080026d8
2                TWO                 0x080026c8
1                ONE                 0x080026b8
0                ZERO                0x080026a8
CLRBITS          CLRBITS             0x08001c6c
```

Appendix: :NONAME Internal Routines

These glossary-documented routines have no dictionary header; their names are CFA labels in the symbol file, for use with DUMP and EXECUTE.

This is a partial list of FISH internal routines in FLASH with the FISH kernel. A separate file may be supplied with more information. This is not complete as some routines are tied to internal conventions. The routines listed have no Name Field Address (NFA) or Link Field Address (LFA) and therefore are not seen in the FISH dictionary when viewed by WORDS.
The names in BOLD in this section are to be found in the symbol table file. Associated with each name will be a number. That number is the CFA address of the routine, for use with DUMP and EXECUTE.
To use a :NONAME word look up it's CFA address in the .sym file, for example BELL is listed in the .sym file as follows: (Check your .sym file for the address!!)
000037F0 t BELL
To us the above information to make a words for use in FISH do as follows:
: Bell 37F0h EXECUTE ; (BELL could be used without conflict)
:NONAME_names are in BOLD TYPE in this section.
Format of this section:
noNFAname (--) As seen in .sym file.
BELL (--) EMIT 07h = terminal BELL sound.
BSOUT (--) EMIT 08h = terminal Backspace.
SIGNON (--) print signon message.
DICTSPACE (-- n) calculate and push the available dictionary space.
RVSPACE (-- n) calculate and push RAM VARIABLE's available space.
FPADDR (-- addr) contains addr of next free flash page.
FPCURR (-- addr) contains flash_save's flash page DP.
FPUSER (-- addr) contains flash_save's flash page UP. (VARIABLE's)
LIT (-- n) compiler internal
Within a colon-definition, LIT is automatically compiled before each 32bit literal number encountered in input text. Later execution of LIT causes the contents of the next dictionary address (the LITERAL item) to be pushed to the stack.
See LITERAL
BRAN (--) COMPILER INTERNAL
The run-time procedure to unconditionally branch. An in-line offset is added to the interpretive pointer IP to branch ahead or back. Compiled by ELSE, AGAIN,
REPEAT.
ZBRAN (f --) COMPILER INTERNAL
The run-time procedure to conditionally branch. If f is false (zero), the following in-line parameter is added to the interpreter pointer to branch ahead or back. Compiled by IF, UNTIL, and WHILE.
PDOTQ (--) COMPILER INTERNAL
The run-time procedure compiled by ." which transmits the following in-line text to the output device.
See ."
XPLOOP (n --) COMPILER INTERNAL
The run-time procedure compiled by +LOOP, which increments the loop index by n and tests for loop completion.
See +LOOP
XDO (Limit Index --) COMPILER INTERNAL
(Limit = addr+cnt Index = addr --)
The run-time procedure compiled by DO which moves the loop control parameters to the return stack.
See DO
XLOOP (--) COMPILER INTERNAL
The run-time procedure compiled by LOOP which increments the loop index and tests for loop completion.
See LOOP

PNUMBER (0 0 addr1 -- d addr2)
(0 0 addr1 -- LSW MSW addr2=addr1+chars)
Convert the ASCII text beginning at addr1+l with regard to BASE into a 64-bit unsigned number d. addr2 is the address of the first un-convertible digit, usually a space (20h) or NULL. Used by NUMBER.
PFIND (addr1 addr2 -- pfa b tf) (ok)
(addr1 addr2 -- ff) (bad)
Searches the dictionary starting at the name field address addr2, matching to the text at addr1. Returns the Parameter Field Address (PFA), length byte (+80h) of the name field, and a Boolean true flag for a good match. If no match is found, only a Boolean false flag is left.
See DFIND
;CREATE (--) Error handler during word compilation.
Used in ; CONSTANT and VARIABLE to reset CSDP System VARIABLE which is used to support a feature called Auto Forget, that forgets words that have errors during compilation, reclaiming dictionary space.
GOTO (cfa --) Used in DOES>
Redirect threaded execution to CFA in another hi-level word.
>>> CAUTION!!! GOTO is a system word, that is exposed only for use by the bold and brave! Use it only at your own risk. The stack and return stack must be in the same level of nesting where you jump from to where you go to!
NEXT (--)
This is the inner interpreter that uses the interpretive pointer IP to execute compiled FISH definitions. It is not directly executed. It jumps to the address pointed to by the address pointed to by the 'W' CPU register. W points to the code field of a definition which contains the address of the code which executes for that definition. This usage of indirect threaded code is a major contributor to the power, portability, and extensibility of FISH. Locations of IP and W are computer-specific.
QEXEC (--)
Issue an error message if not executing.
QPAIR (n1 n2 --)
Issue an error message if n1 does not equal n2. A message is issued when compiler conditionals do not match.
QSTACK (--)
Issue an error message and ABORT if the stack is out of bounds according to the value in INITSO.
BACK (addr --) ALIGNED
Calculate the backward branch offset from HERE to addr and compile it
into the next available dictionary memory address.
CSP_SV (-- addr of CSP)
A system variable temporarily storing the stack pointer position in CSP, for compilation error checking.
HLD_SV (-- addr of NHLD)
A system variable that holds the address of the latest character of text created during numeric output conversion.
INTERPRET (--)
The outer text interpreter which sequentially executes or compiles words or numbers from the input stream depending on the system variable STATE. If the input text cannot be found after searching the dictionary, starting at CURRENT, conversion to a number according to the current base is attempted. If successful, a 32bit number will be placed on the stack. If a decimal point is found in the number, a double number will be placed on the stack, and the decimal point position is saved in DPL, until another number is entered. If input is not a word or number, an error message will issued and ABORT executed.
See DFIND WORD NUMBER DPL and NULL
QUIT (--)
Clears the return stack, stops compilation, and returns control to the interpreter,
INTER. No message is given.

INITRO_SV (-- addr of INITRO)
Return the address of the system variable that contains the initial value of the return stack pointer, INITRO.
See RPSTO
RPSTO (--)
Initialize the return stack pointer from system variable INITR0. Pronounced (RP-store).
INITSO_SV (-- addr of INITSO)
Return the address of the system variable that contains the initial value for the parameter stack pointer, INITSO.
See SPSTO
SPSTO (--)
Initialize the stack pointer from the system variable INITSO. Pronounced "SP-store".
TOGGLE (addr b --)
Complement (FLIP) the byte-contents of addr by the bit-pattern b.
TRAVERSE (addr1 n -- addr2)
Move across a variable-length Name Field in the dictionary. addr1 is the address of either the length byte, or the last letter. If n=1, the motion is toward NFA/hi memory. If n=-l, the motion is toward CFA/low memory. The addr2 resulting, is the address of the other end of the name.
These are not CFA labels:
TXRDY_SUBR:
XOFF_SUBR:
XON_SUBR:
Appendix: FISH differences from other Forths:
Not supported:
(;CODE) PSCOD
USER, VOC-LINK, CONTEXT, VOCABULARY, DEFINITIONS, and all BLOCK/SCREEN SUPPORT.
Renamed words:
MINUS is now NEGATE
DMINUS is now DNEGATE
VLIST is now WORDS
?TERMINAL is now ?KEY
VAR and VARIABLE are both defined.
CON and CONSTANT are both defined.
Modified words:
KEY EMIT's XON to assure input is available.
EXECUTE EMIT's XOFF to allow for long winded words!
: error handling of failed word compilation resets DP to the end of last valid word defined, reclaiming space.
<BUILDS ... DOES> is the same as CREATE … DOES>